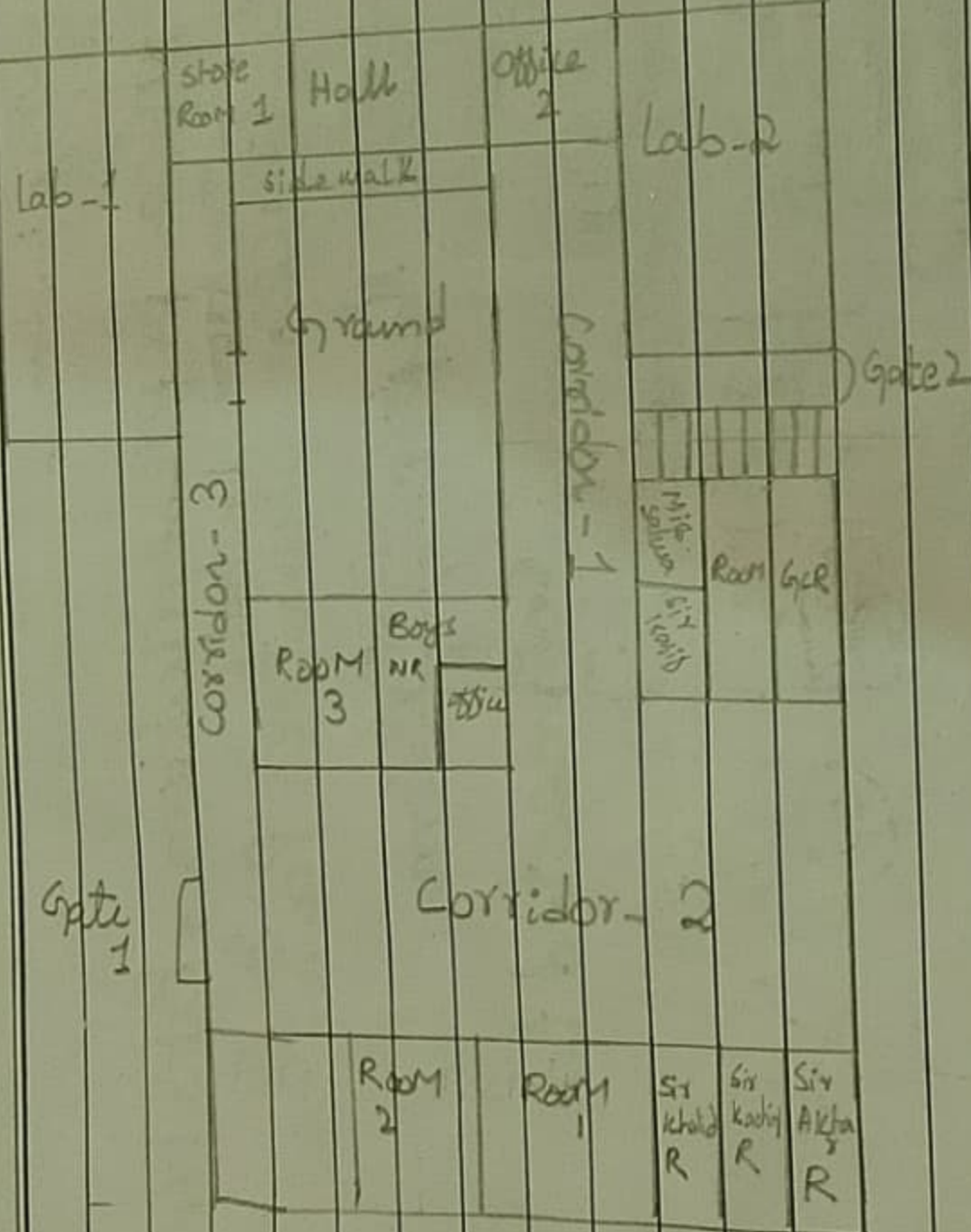
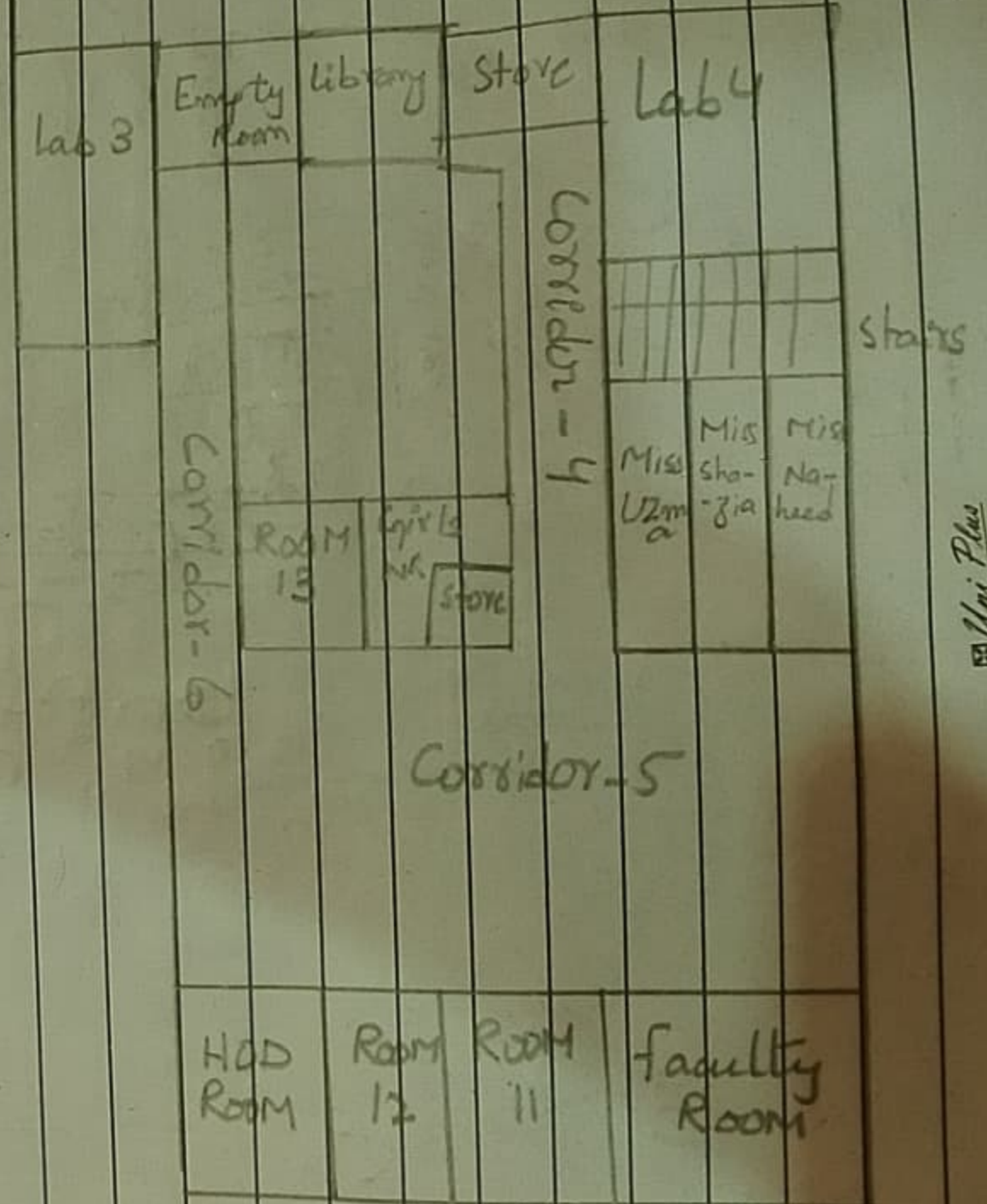


Ground floor



First floor



Q¹ Mapping :

Q² Graphical Representation, BFSTree

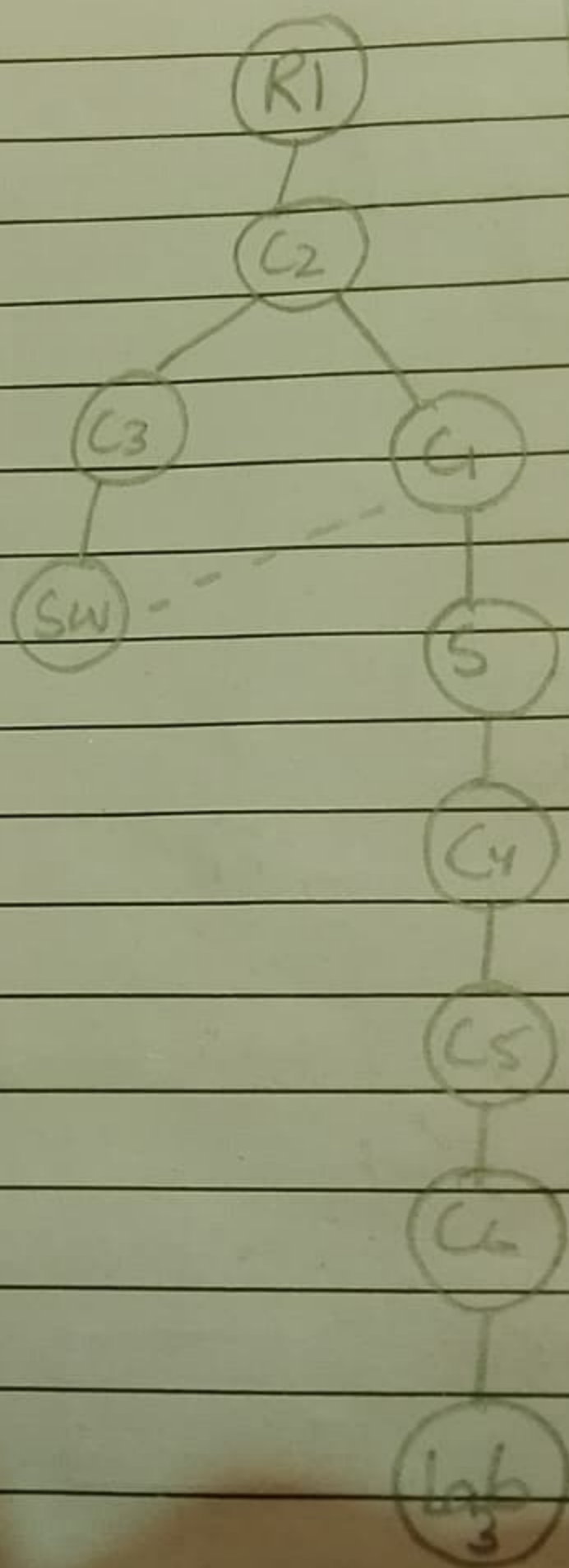
Q³ BFS Trace, Shortest Path

Graphical Representation

Nodes	Connected to
Room 1 (R1)	C ₂
C ₂	C ₃ , C ₁ , R1
C ₃	C ₂ , SW
C ₁	SW, C ₂ , S
SW	C ₃ , C ₁
S (stairs)	C ₁ , C ₄
C ₄	S, C ₅
C ₅	C ₄ , C ₆
C ₆	C ₅ , Lab 3
Lab 3	C ₆

BFS Tree :

Tree



BFS Trace =

Step 0 (Initialization)

Q: { R₁ }

V: { R₁ }

Dequeue: R₁

Q: { C₂ }

V: { R₁, C₂ }

Dequeue: C₂

Q: { C₃, C₁ }

V: { R₁, C₂, C₃, C₁ }

Dequeue C₃

Q: { C₁, SW }

V: { R₁, C₂, C₃, C₁, SW }

Dequeue: C₁

Q: { SW, S }

V: { R₁, C₂, C₃, C₁, SW, S }

Dequeue SW

Q: { S }

V: { R₁, C₂, C₃, C₁, SW, S }

Dequeue S
 Q: { C4 }
 V: { R1, C2, C3, C1, SW, S, C4 }

Dequeue C4
 Q: { C5 }
 V: { R1, C2, C3, C1, SW, S, C4, C5 }

Dequeue C5
 Q: { C6 }
 V: { R1, C1, C2, C3, C1, SW, S, C4, C5, C6 }

Dequeue C6
 Q: { Lab3 }
 V: { R1, C1, C2, C3, C1, SW, S, C4, C5, C6, Lab3 }

Dequeue Lab3
 Q: { Destination }
 V: { R1, C1, C2, C3, C1, SW, S, C4, C5, C6, Lab3 }

Shortest Path:

R1 → C2 → C1 → S → C4 → C5 → C6 → Lab3

[illegible]

Route: 1

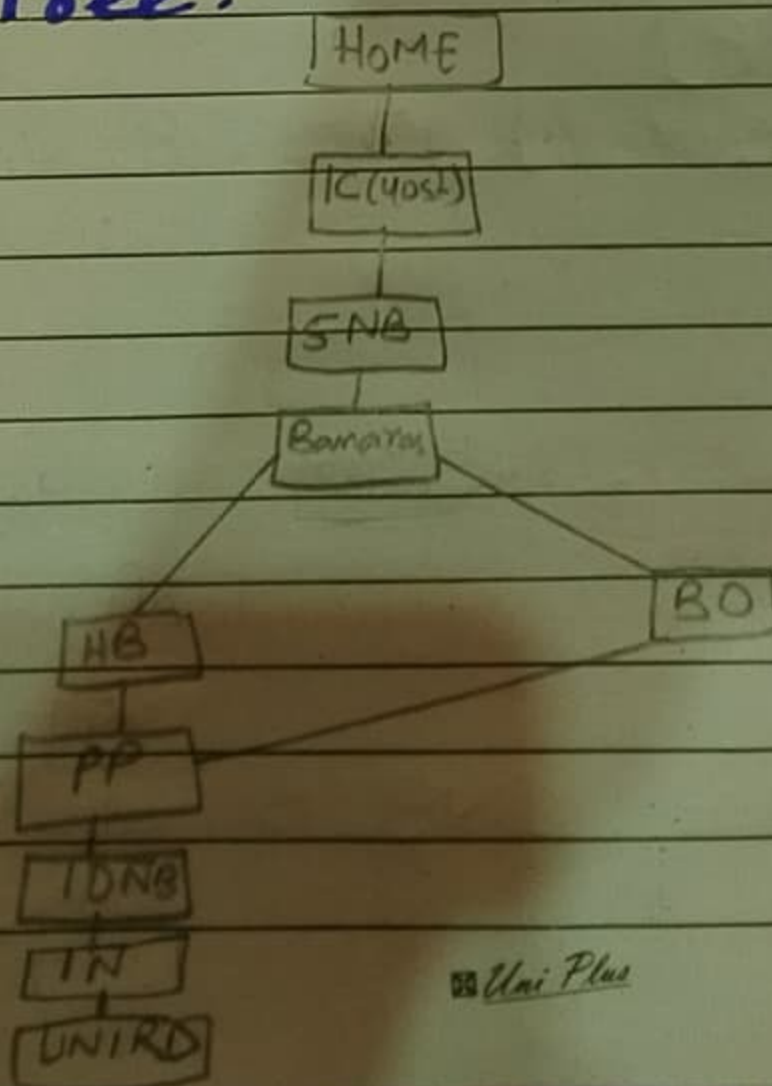
$$0.5 + 1.95 + 1.65 + 1.90 + 0.14 + 1.67 + 0.08 + 0.09 + 0.10 = 8.08 \text{ KM}$$

Route: 2

$$0.5 + 1.95 + 1.65 + 2.50 + 1.51 + 0.08 + 0.09 + 0.10 = 8.38 \text{ KM}$$

Nodes	HOME	Connected to
Islamchik (40st)		HOME, SNB
SNB		IC (40st), Banaras
Banaras		HB, BO
Habib Bank (HB)		PP, BD
Board Office (BO)		PP
Petrol Pump (PP)		10 Num (Liaquatabad)
Liaquatabad (10NB)		IN
Isa Nagri (IN)		Uni road
UNI Road		Destination

BFS Tree:



Step: 0 Initialization

Date 20
M T W T F S S

Q: [Home]
V: [Home]

Dequeue: Home

Q: [IC(40st)]
V: [Home, IC(40st)]

Dequeue: IC(40st)

Q: [SNB]
V: [Home, IC(40st), SNB]

Dequeue: SNB

Q: [Banaras]
V: [Home, IC(40st), SNB, Banaras]

Dequeue: Banaras

Q: [HB, BO]
V: [Home, IC(40st), SNB, Banaras, HB, BO]

Dequeue: HB

Q: [BO, PP]
V: [Home, IC(40st), SNB, Banaras, HB, BO, PP]

Dequeue: BO

Q: [PP]
V: [Home, IC(40st), SNB, Banaras, HB, BO, PP]

Dequeue: PP

Q: [IONum]
V: [Home, IC(40st), SNB, Banaras, HB, BO, PP, IONum]

Dequeue IONB

Q: [Isa Nagri]

V: [Home, IC, SNB, Banaras, HB, BO, PP, IONB, IN]

Dequeue Isa Nagri

Q: [Uniroad]

V: [Home, IC (40st), Banaras, HB, BO, PP, IONB, IN, UNI Road]

Dequeue UNI Road

Q: [Destination]

V: [Home, IC (40st), SNB, Banaras, HB, BO, PP, IONB, IN, Uniroad]

Shortest Path

HOME → IC(40st) → SNB → Banaras → HB → PP → IONB →
 ISA Nagri → Uniroad

Alternate Path

HOME → IC(40st) → SNB → Banaras → BO → HB → PP →
 IONB → Isa Nagri → Uniroad